Fourier Qualia Wavescapes: Hierarchical Analyses of Set Class Quality and Ambiguity

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We introduce a novel perspective on set-class analysis combining the DFT magnitudes with the music visualisation technique of wavescapes. With such a combination, we create a visual representation of a piece's multidimensional qualia, where different colours indicate saliency in chromaticity, dyadicity, triadicity, octatonicity, diatonicity, and whole-tone quality. At the centre of our methods are: 1) the formal definition of the *Fourier Qualia Space* (FQS), 2) its particular ordering of DFT coefficients that delineate regions linked to different musical aesthetics, and 3) the mapping of such regions into a coloured wavescape. Furthermore, we demonstrate the intrinsic capability of the FQS to express qualia ambiguity and map it into a synopsis wavescape. Finally, we showcase the application of our methods by presenting a few analytical remarks on Bach's *Three-part Invention* BWV 795, Debussy's *Reflets dans l'eau*, and Webern's *Four Pieces for Violin and Piano*, Op. 7, No. 1, unveiling increasingly ambiguous wavescapes.

Keywords: *Fourier Qualia Space*; wavescapes; Fourier transform; music analysis; harmonic qualities.

1 INTRO: SHEDDING LIGHT ON WAVESCAPES AND THE DFT

A key question for tonal music analysis is how to connect the interrelations of tonal structures present at different timescales and how harmonic patterns, modulation plans, and key regions interact at local, regional, and global levels. Understanding the hierarchical representation of harmonic structure in Western classical music is thus a central aspect of music analysis. It provides insights into the organisation of tonal and harmonic relationships from the surface pitch and chordal elements (Bharucha and Krumhansl 1983; Granroth-Wilding 2013; Krumhansl and Kessler 1982; Rohrmeier 2011). These dynamic structural changes have been pursued computationally to foster music analysis, with a strong emphasis on empirical and corpus studies, namely to assess the tonal organisation as intuitive visual and spatial representations at multiple hierarchical levels.

1.1 HIERARCHICAL MUSIC ANALYSIS WITH KEYSCAPES

Keyscapes are computational visual representations of the tonal structure of a piece of music. They commonly map maximum-likelihood estimates of a key-finding algorithm to musical segments at different temporal scales—from microelements on the musical surface to macro inference of a piece's global key (Sapp 2001; Sapp 2005).

Thus, keyscapes show the hierarchical organisation of tonality using a colour-coded scheme that indicates the degree of familiarity with a key. Its use is suitable for music analysis as it can intuitively unveil patterns of modulation, tonicisation, and key relations in a piece and highlight overall tonal coherence or diversity of

musical structure. Furthermore, it fosters new methods for empirical music analysis at scale (Dominguez 2009) and applies symbolic and audio music encodings. A generalisation of keyscapes rendering probabilistically weighted clusterings of key estimates for each segment instead of a single maximum value has been explored in Robert Lieck and Martin Rohrmeier's (2020) work, which entail some tonal ambiguity concerning uncertainty in the key-finding procedure.

1.2 ANALYSING PITCH-CLASS SETS WITH THE DFT

Independently of the development of keyscapes, recent years have seen a growth of music-theoretical and analytical approaches drawing on the capacities of the discrete Fourier transform (DFT) for music analysis based on pitch-class sets (pcsets). The existing literature has proposed scalable and music-theoretically interpretable computational methods for analysing the hierarchical structure of pitch-class relations in empirical corpus studies. In essence, the DFT, initially introduced by David Lewin (1959) and recently revitalised (Amiot 2016; Bernardes et al. 2023; Bernardes et al. 2016; Harding 2020; Quinn 2006, 2007), decomposes pitch-class distributions into components resonating with prototypes of music-theoretically relevant structures. The DFT is agnostic to traditional tonal notions, making it suitable for analysing styles beyond common-practice tonality.

While applying the DFT to specific pcsets has been explored from various perspectives, several approaches have been used to analyse segments of pieces or entire compositions in corpus studies. A straightforward application is to compute pcsets of music segments with multiple granularity levels (Harding 2020; Yust 2019) This, however, requires choosing a particular level of analysis, which is challenging to do in a principled way without intricate knowledge of a given piece that can only be obtained through detailed manual analysis.

To avoid the issue of choosing specific levels of partitioning a piece, a different approach seeks *exhaustive segmentation*, where each possible contiguous segment of a piece is considered, leading to a bottom-up hierarchical structure. Moreover, the core assumption of the keyscapes method requires the existence of major and minor keys, thus strongly limiting its scope of meaningful applicability beyond tonal music.

1.3 COMBINING KEYSCAPES AND THE DFT: WAVESCAPES

To overcome these limitations, *wavescapes* (Viaccoz 2023) have been recently proposed as a novel visualisation technique that combines the advantages of keyscapes visualisation with the power and flexibility of DFT-based music analysis. Wavescapes are based on the idea that each Fourier coefficient $\hat{f}_k$ resonates with a specific prototype of music-theoretically relevant pitch-class structures (Amiot 2016). The prototypes are interpreted as the following qualities: chromaticity ($\hat{f}_1$), dyadicity ($\hat{f}_2$), triadicity ($\hat{f}_3$), octatonicity ($\hat{f}_4$), diatonicity ($\hat{f}_5$), and whole-tone quality ($\hat{f}_6$). Given a pitch-class distribution, the magnitude of each DFT coefficient indicates, loosely speaking, how the distributionis representative of the corresponding prototype (Amiot 2016; Chiu 2021; Yust 2016, 2019). Wavescapes can unveil the patterns of tonal stability and tension, modulation, and similarity within a piece or a corpus by highlighting and quantifying degrees of tonal coherence or diversity. Practically, wavescapes can be produced using the dedicated Python library

`wavescapes`[1] or the web application `midiVERTO`[2] (Harasim 2022; Moss 2022), which does not require programming skills and allows for interactive analytical exploration.

The DFT of pcsets yields a 12-dimensional vector of complex numbers. Typically, only the non-trivial dimensions, i.e., coefficients 1 through 6, are considered due to symmetry (Amiot 2016). Consequently, DFT data is essentially multi-dimensional, although analysts often opt to display the dimensions separately or select only a subset. Interpreting six-dimensional (6-D) spaces can be challenging; hence, exploring methods to reduce this complexity is a logical consideration. A recent approach to reduce the dimensionality of wavescapes is the *summary wavescape*, used in Sabrina Laneve et al.'s (2023) article to study the diachronic development of Debussy's musical style. The summary wavescape only stores information on the most salient Fourier coefficient at each time scale. It can thus reduce the complexity and redundancy of the original 6-D wavescape and provide a clearer and more concise overview of the tonal hierarchy of a piece at one glance. However, this approach has no explicit notion of Fourier coefficient ambiguity as it only shows the most salient one for a given segment at a particular time scale.[3]

2 THE *FOURIER QUALIA SPACE*: DEFINITION AND PROPERTIES

The *Fourier Qualia Space* (FQS) is a hexagonal plane partitioned into six regions, each corresponding to distinct harmonic qualities derived from the DFT (Pereira 2025). Each unique point within this space denotes set classes, and their coordinates are intricately linked to their coefficient magnitudes. In other words, all pcsets belonging to the same set class exhibit identical geometric coordinates irrespective of the pitch-classes within each set. Below, we comprehensively explore the mathematical and geometrical underpinnings of the FQS, elucidating hierarchical aspects of harmonic qualia and degrees of qualia ambiguity.

2.1 PROJECTING SET CLASSES IN THE FQS

The FQS uses the *RadViz* dimensionality reduction technique (Dzemyda 2013) to project the six-dimensional non-trivial DFT magnitudes of a set class onto a two-dimensional hexagon with $n = 6$ anchors positioned equidistantly, as illustrated in Figure 1. The anchors $s_j = (x_j, y_j)$ with $j = 1, \dots, n$ are the hexagon vertices and represent the DFT coefficients and their respective harmonic qualities. A physical model of springs is used to determine the two-dimensional coordinates of a set class. The DFT coefficient magnitudes $|\hat{f}_1|, \dots, |\hat{f}_6|$ correspond to the stiffness of the n springs connected to the anchors S_n. Given the DFT magnitudes of a set class $K = (|\hat{f}_1|, \dots, |\hat{f}_6|)$, the final position of K in the FQS is the two-dimensional point $P_K = (x, y)$ where the sum of the spring forces is null, such that:

[1] Link to the wavescapes library: https://github.com/DCMLab/wavescapes.

[2] Link to the midiVERTO application: https://dcmlab.github.io/midiVERTO.

[3] Ambiguity has a rich history in musical composition and analysis. For example, it is associated with tonality, where identical pitches can be interpreted differently in various modes or tonalities. The qualia ambiguity we seek to discuss aligns with Temperley's (2007) concept, i.e., indicating the degree to which a set class is linked with two or more harmonic qualities.

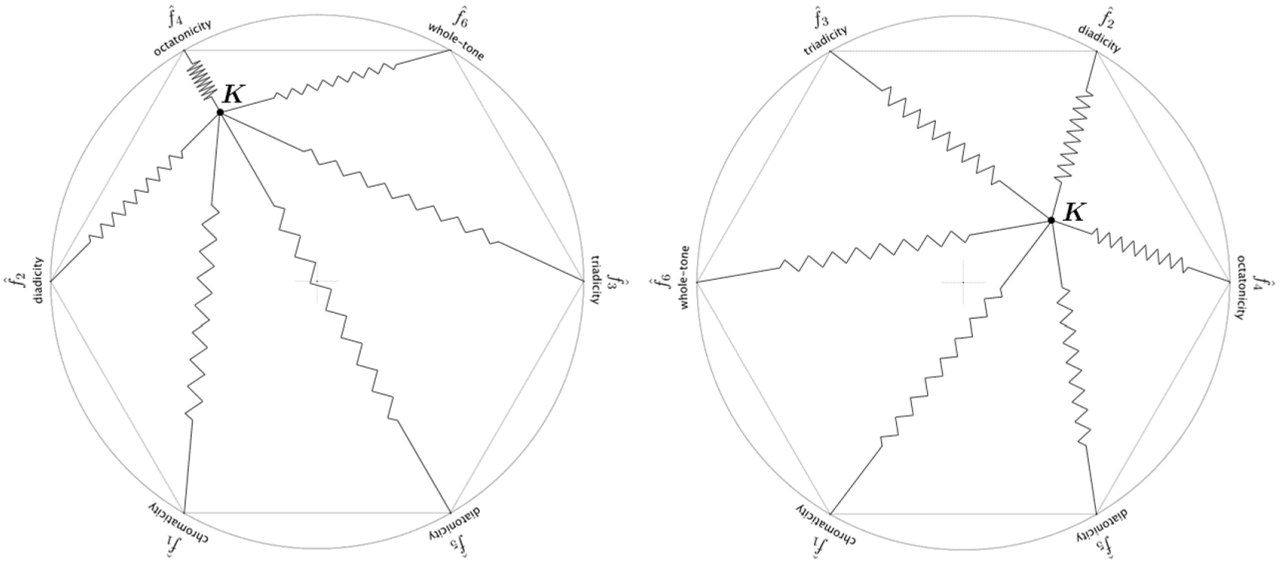


FIGURE 1. Two *RadViz*-derived hexagonal depictions for the DFT coefficients. Left and right images differ in the ordering of coefficients, resulting in distinct coordinates for Stravinsky's *Petrushka* chord K (two major triads a tritone apart).

$$P_K = \left(\frac{\sum_{j=1}^{n} x_j \cdot \left|\hat{f}_j\right|}{\sum_{j=1}^{n} \left|\hat{f}_j\right|}, \frac{\sum_{j=1}^{n} y_j \cdot \left|\hat{f}_j\right|}{\sum_{j=1}^{n} \left|\hat{f}_j\right|} \right). \tag{1}$$

Figure 1 illustrates how the hexagon vertices are associated with the DFT coefficients. It also displays the position of pcset K determined with equation (1)—K is the *Petrushka* chord with pitch-classes {0,1,4,6,7,10}, belonging to set class (013679)—, exhibiting DFT magnitude values $\left|\hat{f}_1\right| = 0$, $\left|\hat{f}_2\right| = 2$, $\left|\hat{f}_3\right| = 0$, $\left|\hat{f}_4\right| \approx 3.5$, $\left|\hat{f}_5\right| = 0$, $\left|\hat{f}_6\right| = 2$, which determine the stiffness of each spring and K's position in space.

Mapping the DFT coefficient magnitudes to the anchors $S_1, S_2, \ldots, S_6$ in the FQS is crucial to the analytical interpretation of a set class position, since distinct arrangements result in different positionings to a given set class, as Figure 1 shows. To adress this issue, we adopt the order reported in Samuel Pereira et al.'s (2025) work, which empirically evaluated the order that best: (1) promotes a wider dispersion of pcsets in the FQS and (2) maintains the highest possible correlation between pcsets' positioning in the FQS and in the six-dimensional DFT magnitude space. The permutation adopted resulted in the following order $\hat{f}_1, \hat{f}_5, \hat{f}_3, \hat{f}_4, \hat{f}_6, \hat{f}_2$.[4] Although other methods could be used to define the order of the DFT coefficients on the *RadViz* circle, the current order facilitates several analytical and interpretive properties that we detail below.

[4] The order described was determined by computing Euclidean distances between all $\mathbb{Z}_{12}$ set classes in both the six-dimensional space and all possible anchor permutations in the FQS, correlating both matrices to find the optimal permutation. The order $\hat{f}_1, \hat{f}_5, \hat{f}_3, \hat{f}_4, \hat{f}_6, \hat{f}_2$ maximised the Pearson correlation coefficient with $r = .72$.

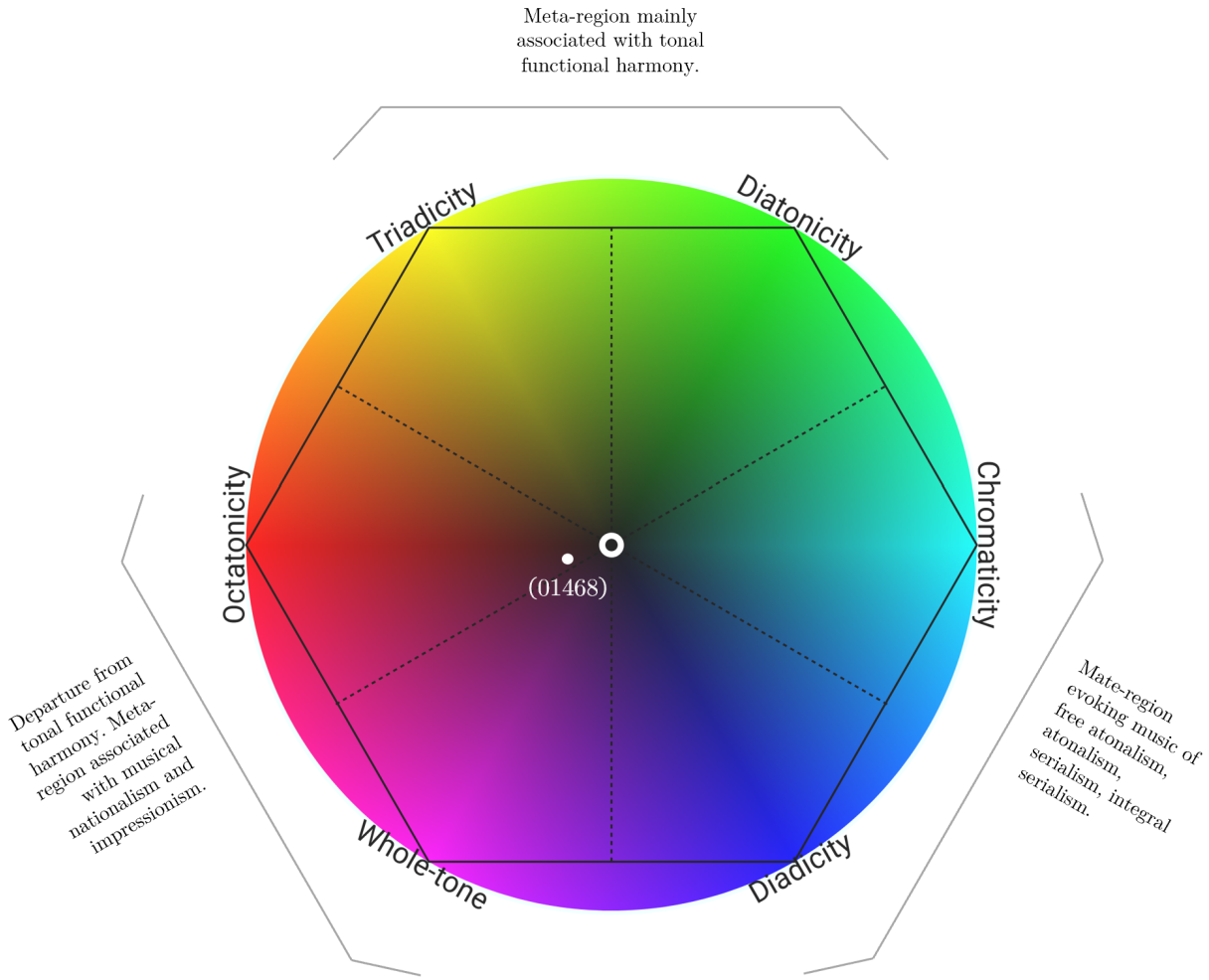


FIGURE 2. Depiction of the FQS and the final arrangement of DFT coefficients (Pereira 2025). The image also illustrates the three meta-regions arising from this specific order.

2.2 PROPERTIES: (FUZZY) REGIONS, CONVEXITY, AND "SINGULARITY"

The adopted coefficient order has proven pivotal to musical analysis. This value is understood by delineating six regions, each representing higher saliency in its respective quality, as illustrated in Figure 2. Furthermore, this allows the identification of three meta-harmonic regions, each encompassing two harmonic qualities: (1) diatonicity ($\hat{f}_5$) and triadicity ($\hat{f}_3$); (2) octatonicity ($\hat{f}_4$) and whole-tone quality ($\hat{f}_6$); and 3) dyadicity ($\hat{f}_2$) and chromaticity ($\hat{f}_1$). Meta-regions, illustrated in Figure 2, are linked with different musical aesthetics. The first group is associated with tonal functional harmony; the second group is significant in impressionism and musical nationalism; and the third group is fundamental to 20th-century music and atonal music.[5]

However, any set class in the FQS is a convex combination of the anchors' positions. Therefore, any pcset positioned away from anchors is a convex combination of two or more DFT coefficients. This mathematical

[5] We should also note that the adopted order presents a new analytical layer by dividing the FQS into groups of three DFT coefficients. This allows us to interpret the positioning of a set class based on the the axis that separates $\hat{f}_3, \hat{f}_5, \hat{f}_1$ vs. $\hat{f}_2, \hat{f}_6, \hat{f}_4$ indicating the presence or absence of tritones in a given set class.

feature suggests that the regions should not be seen as confined spaces, motivating us to depict their boundaries with dashed lines, as can be seen in Figure 2. In fact, they allow many convex combinations between the six harmonic qualities, particularly when a set class is near the space's centre. For instance, the pentachord (01468), shown in Figure 2, is near the centre and the boundary between the $\hat{f}_4$ and $\hat{f}_6$ regions. A restricted view might classify this set class as solely octatonic. However, its position is a convex combination of all harmonic qualities, with a DFT magnitude profile of $\left|\hat{f}_1\right| = 0.52$, $\left|\hat{f}_2\right| = 1.73$, $\left|\hat{f}_3\right| = 2.23$, $\left|\hat{f}_4\right| = 1$, $\left|\hat{f}_5\right| = 1.93$, $\left|\hat{f}_6\right| = 3$. The greater saliency in $\hat{f}_6$ plots this pcset towards the whole-tone area, but the secondary saliency in $\hat{f}_3$ shifts its position towards triadicity, resulting in a final position inside the $\hat{f}_4$ region. This effect is more apparent when the DFT magnitude profile of a pcset does not have a very prominent primary quality and is positioned near the centre. This property of the FQS expands existing reductionist approaches to the six DFT coefficients by encompassing their interaction, as opposed, e.g., to Sabrina Laneve et al.'s (2023) approach that retain the most salient coefficient only.

The pentachord (01468) exemplifies that the FQS centre is the peak of qualia ambiguity, implying a balance of forces between two or more DFT coefficients. Hence, the FQS centre and its vicinity form a "singular" zone due to its qualia ambiguity, with two or more qualities residing with equal intensity. Consequently, a set classes' proximity to the centre broadens its qualia interpretation. Figure 2 illustrates this with a gradient that darkens towards the centre, symbolising increasing qualia ambiguity. Despite its peculiarity, the ambiguous zone in space is beneficial as it allows the assessment of the recurring analytical remark of ambiguity in music. Furthermore, Figure 2 illustrates how we coloured the FQS to map the set-class position's colour into a wavescape. In our approach, the six harmonic qualities are represented by a unique colour in a radial gradient.

3 QUALIA HIERARCHIES IN BACH, DEBUSSY, AND WEBERN

We aim to elucidate the harmonic qualities of three pieces by composers of different historical periods: Bach's *Three-part Invention* in F minor, BWV 795; Debussy's *Reflets dans l'eau* (Images, Book I, No. 1); and Webern's *Four Pieces for Violin and Piano*, Op. 7, No. 1. Our analysis focus on their respective wavescapes and positioning within the FQS, illustrating the relationship between the meta-regions and the pitch-organisation across different compositional periods.

Bach's *Three-part Invention* BWV 795 in F minor unfolds through two themes exhibited in Figure 3a. The first motif involves an ascending leap of a third followed by a descending second, expanded to include a tritone leap and a descent from F to A♭. The second motif, a *lament bass*, involves a chromatic descent from the tonic to the dominant, followed by a cadential ornamentation. These motifs undergo inversion, expansion, imitation, and combination throughout the piece. The tonal plan of the composition evolves through subtle modulations, creating a sense of tonal unpredictability with cadences in C minor, A♭ major, E♭ major, D♭ major, and F minor. Despite the frequent emergence of chromatic elements, the pitch organisation favours diatonic structures, resulting in low variability in the work's qualia. This is reflected in the wavescape illustrated in Figure 3b, where the prevalence of diatonicity ($\hat{f}_5$) is visible, as a consequence of the musical

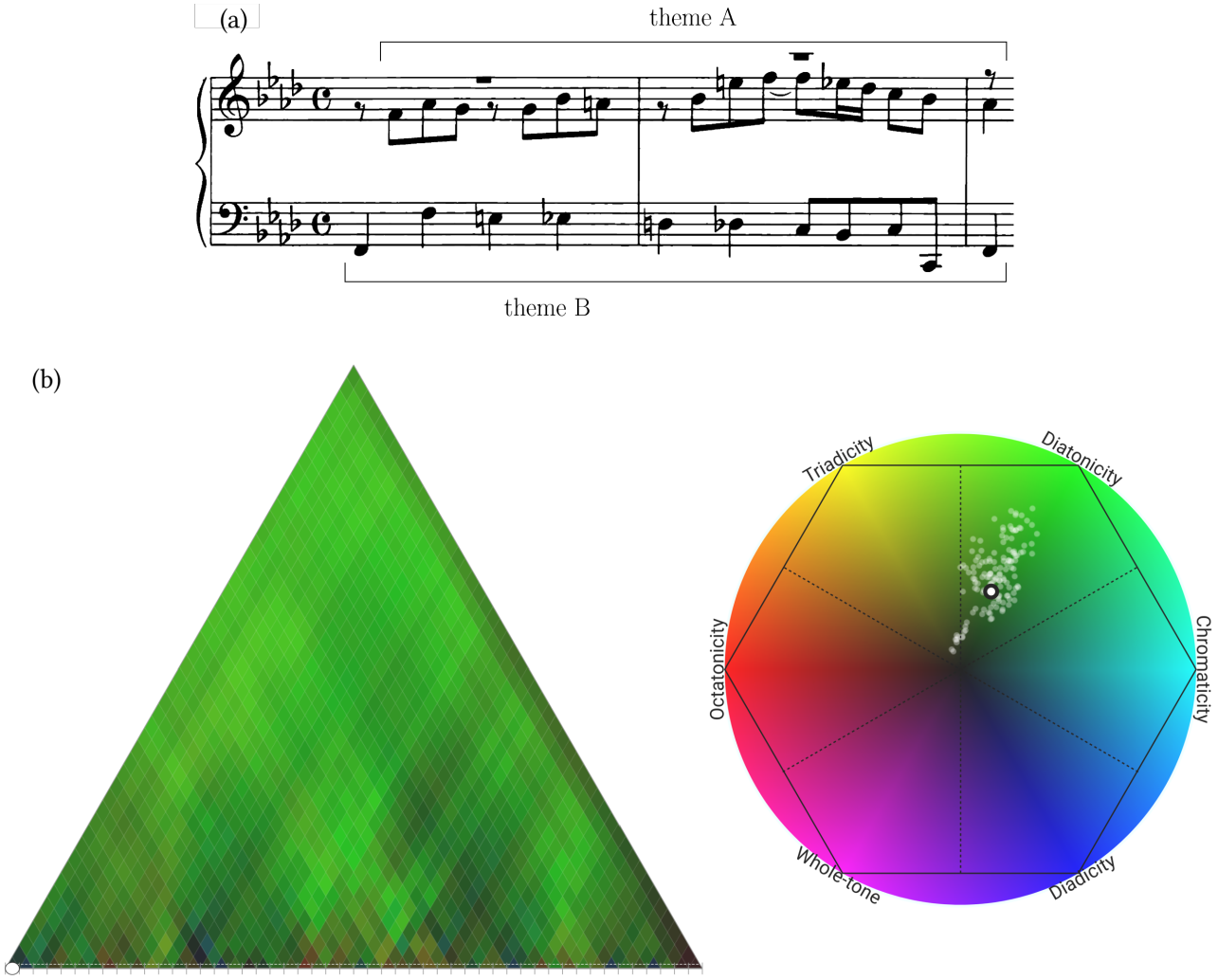


FIGURE 3. In 3a are mm. 1 and 2 of Bach's BWV 795, highlighting the two structural motifs in the piece (score in public domain). In 3b one can examine both the wavescape and the FQS positioning of the piece's segments—time resolution (tr) = ♩ and window length (wl) = 26.

segments' positioning in the FQS lying well within the diatonic region and meta-region of functional harmony. The overall brightness of the wavescape suggests low qualia ambiguity. However, focal points of ambiguity are associated with lower hierarchical levels of the composition. As we ascend in the wavescape, diatonicity becomes increasingly apparent. Despite the tonally ambiguous sensation imparted by the structuring use of the chromatic motif, the prevalence of diatonic pitch structures and clear cadences in various tonalities attest to diatonicity as the primary harmonic quality in the piece.

Debussy's Images, Book I, No. 1 exemplifies increased complexity and structural relevance in harmonic qualia as the result of an anchored harmonic path from opposing regions in the FQS. The wavescape of this piece exhibits pronounced colour variance, indicating the prominence of various harmonic qualities, as Figure 4a depicts. Darker patches related to qualia ambiguity emerge at higher hierarchical levels, suggesting a more complex predominant harmonic quality. The wavescape analysis reveals a pendulum-like movement between diatonic (greenish tones) and whole-tone (purplish tonalities) qualities.

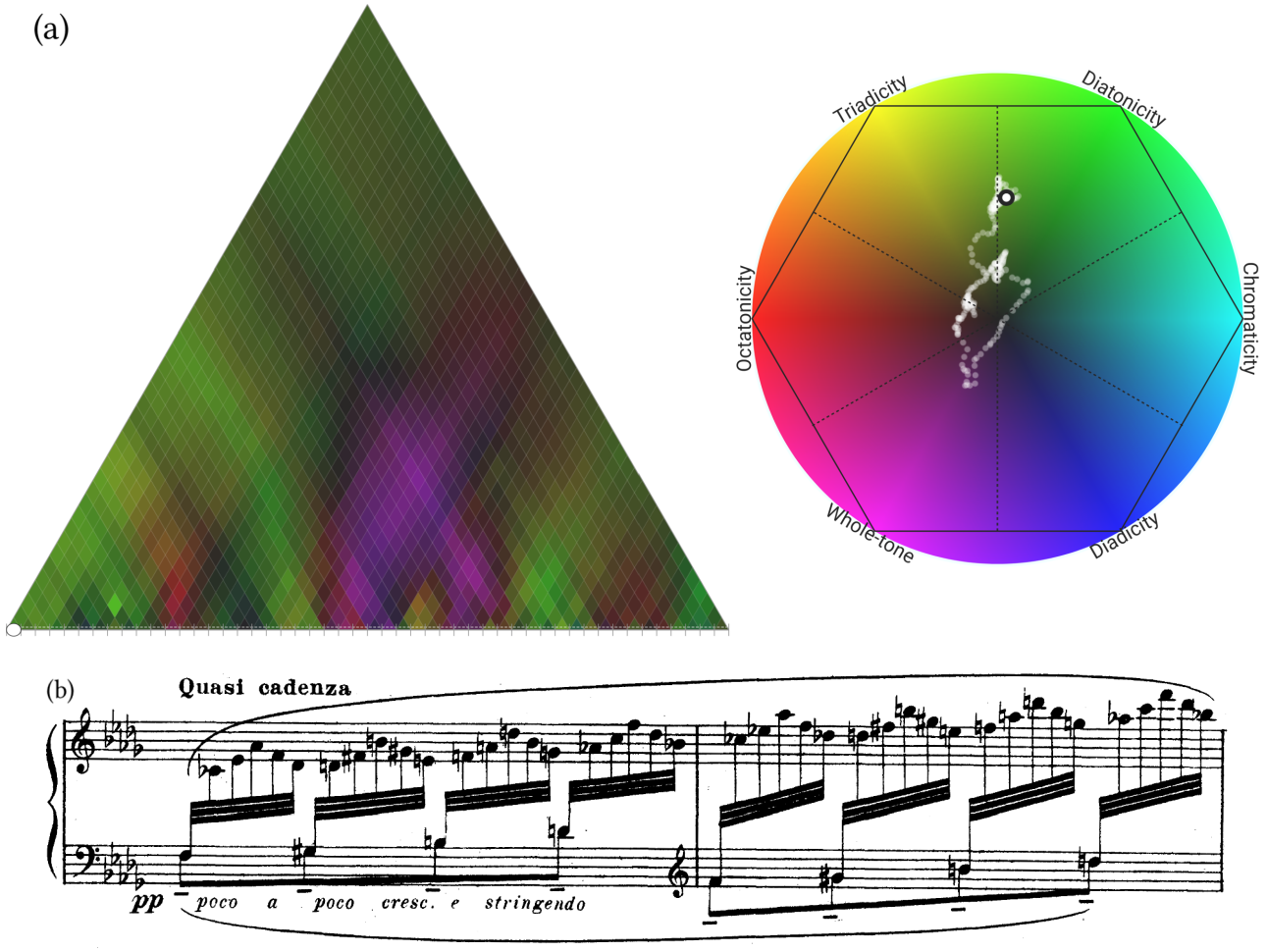


FIGURE 4. In 4a is the wavescape for *Reflets dans l'eau* outlining the movement between $\hat{f}_5$ and $\hat{f}_6$ qualities, as well as its FQS positioning performing a circular trajectory in space—tr = ♩ and wl = 86. In 4b are mm. 20 and 21 with the sequence of ninth chords that create the wavescape's redish spot (score in public domain).

This movement establishes diatonicity as the starting and ending point of the harmonic motion, with whole-tone quality as the contrasting material. This movement is embodied in the arrangement of the musical segments within the FQS, forming a clockwise motion through the meta-region encompassing $\hat{f}_4$ and $\hat{f}_6$. Two pitch organisations favour the DFT coefficient $\hat{f}_5$. The use of the major scale and pentatonic segments, closest to the anchor of $\hat{f}_5$ in the FQS, emphasises diatonicity.[6] The section beginning with the *En animant* passage at measure 44, composed of the whole-tone scale with pitch-classes {1,3,5,7,9,11}, contributes to the prominence of $\hat{f}_6$ and the purplish patch in the wavescape. A brief octatonic section is evident in the lower tiers of the wavescape, associated with the seventh and ninth chords sequence in the *Quasi Cadenza* at measure 20 (this sequence is shown in Figure 4b). The presence of $\hat{f}_4$, while not directly emanating from the auditory experience of the piece, links to the fact that four notes out of the five forming a major seventh and

[6] The DFT coefficient $\hat{f}_5$ also tends to exhibit high magnitude for pentatonic sets. This coefficient measures how close the analysed set is to a segment of the circle of fifths.

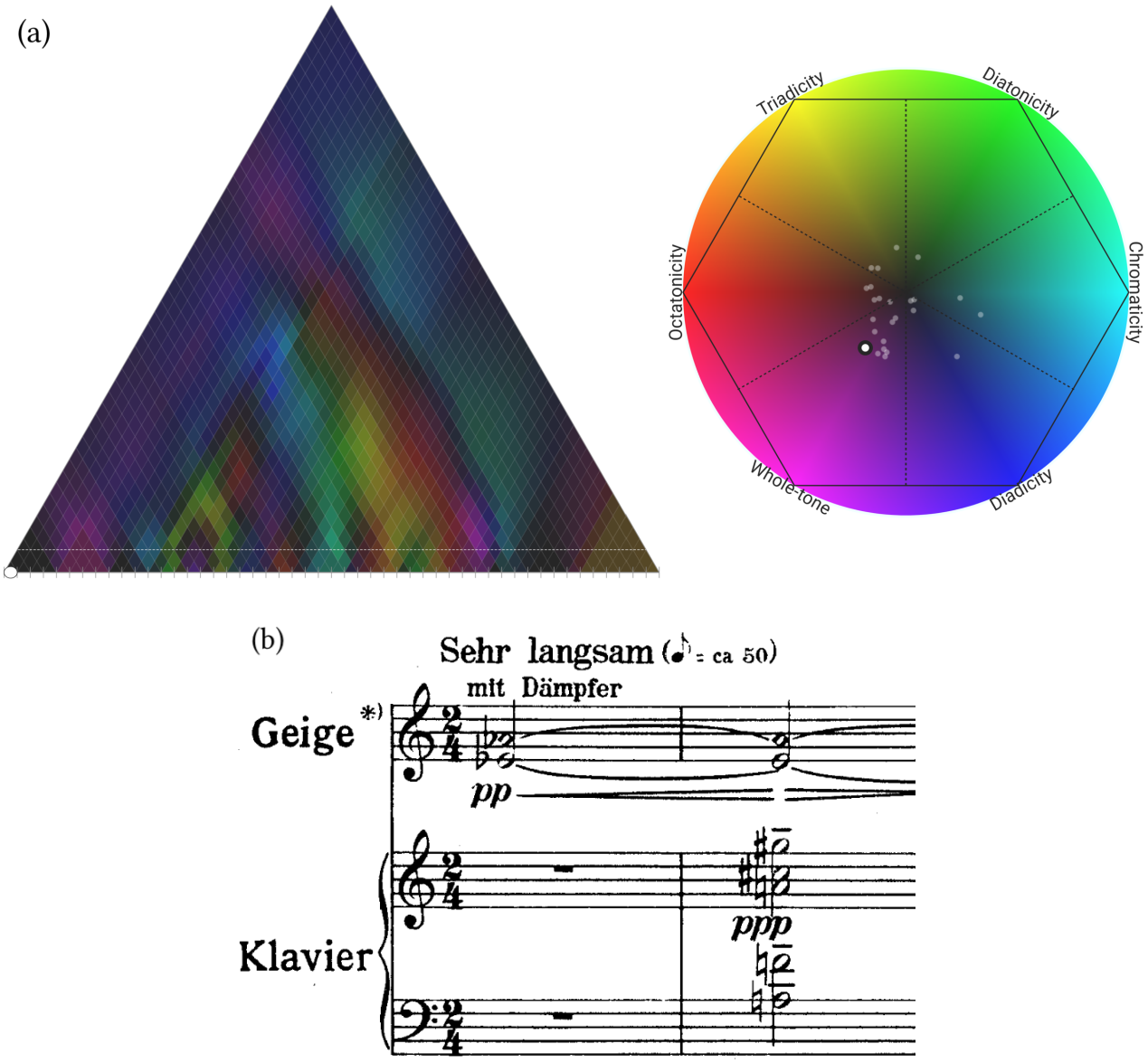


FIGURE 5. In 5a is the two-triangle structure of Webern's Op. 7/1 wavescape and its FQS positioning—tr = ♩ and wl = 8—, and in 5b is the piece's opening chord (score in public domain).

ninth chord constitute a subset of the octatonic scale, and to the fact that the motif is sequenced by minor thirds.

Webern's *Four Pieces for Violin and Piano* (Op. 7, No. 1) is a nine-measure miniature composition. The piece is divided into two sections, each characterised by different roles for the violin and piano. In the first section (mm. 1-5), the violin carries the melody, supported harmonically by the piano. This dynamic is reversed in the second section (mm. 6-9), with the piano taking the melodic lead and the violin providing an *ostinato* accompaniment. The bipartite structure of the work is reflected in its qualia wavescape (in Figure 5a), which reveals two nested triangular structures. These structures suggest a two-fold structure (mm. 1-5 and 6-8), with a subtle cadential coda (second half of m. 8 to m. 9). The coda, marked by a *ritardando* and a thinning texture, signals the conclusion of the phrase. The greenish tones on the right side of the wavescape further emphasise the coda's individuality, indicating diatonicity as a structuring element. This diatonic character arises from

the cadence's explicit reference to a subset of an E♭ major scale (pitch-classes {2,3,7,10}). The final E♭-major chord played by the piano reinforces this diatonic conclusion.

Moreover, the wavescape reveals that the section from mm. 6 to 8 undergoes brief, slight deviations into various harmonic qualities, displaying mostly yellow, green, and red tonalities. These hues arise from a complex texture where each note subtly alters the work's qualia. Conversely, the piece's initial part, represented on the wavescape's left side, shows less qualia variability, with a dominance of bluish tones linked to dyadicity ($\hat{f}_2$) and faint purplish patches related to whole-tone quality ($\hat{f}_6$). The prevalence of $\hat{f}_2$ situates the work in the meta-region connecting $\hat{f}_2$ and $\hat{f}_1$, underscoring the first chord's importance for the work's development and the high qualia ambiguity conveyed by the wavescape. Listening to the second measure's chord depicted in Figure 5b, we perceive it differently based on our attentional focus. Concentrating on the pitch-classes {1,3,5,9} highlights the whole-tone quality, while focusing on the piano's G♯ and the pitch-classes {3,8,9} accentuates the dyadic quality.[7] All these elements suggest a composition of high qualia plasticity, typical of the period, swiftly transitioning between multiple harmonic universes and often highlighting two or more qualities in the same harmonic event, depending on our auditory focus.

4 FINAL REMARKS

We proposed a method combining the FQS—a hexagonal space for interaction of six harmonic qualities—with the hierarchical visualisation technique, wavescapes. This approach visualises a work's qualia in a triangular structure, with colours indicating salience in six harmonic qualities and higher levels offering a more global qualia view. The complementarity between the two representations is noteworthy in light of Saussure's (2011) syntagmatic/paradigmatic dichotomy, which McCreless (1991) employs in the context of the musical discourse: (1) the FQS provides a paradigmatic visualisation by arranging analytical information in a non-temporal way and (2) the wavescape presents the analytical information in a syntagmatic context, ordered in the time of the music itself. Integrating FQS's qualia ambiguity into the wavescape visualisation enhances its interpretation, providing insights into a work's dynamic qualia nature. This is crucial as a musical passage's perception can vary based on its interval and pitch organisation. The wavescapes of the three pieces presented, with their progressively darker tones, imply higher qualia ambiguity. Future application of our method to an extended corpus of scores could elucidate the relationship of qualia ambiguity across compositional periods. In addition, future analysis would benefit from the ability to include multiple orders of DFT coefficients in the FQS to illustrate how they affect the resulting wavescapes. Furthermore, our method allows the identification of distinct spots at high hierarchical levels in some composers, indicating movement between harmonic qualities and their structural importance. Studying these movements and the specific syntactic roles of distinct harmonic qualities could be insightful. Lastly, our analysis of Webern's Op. 7, No. 1

[7] The DFT coefficient $\hat{f}_2$ is more salient for sets arranged in tritone sequences at a minor second distance, as exemplified by the pcset {0,5,6,11}.

reveals its high qualia plasticity, signifying music in constant qualia metamorphosis, navigating various harmonic fields without a discernible destination, meriting further exploration.

REFERENCES

Amiot, E. 2016. *Music Through Fourier Space: Discrete Fourier Transform in Music Theory*. Computational Music Science. Springer.

Bernardes, G., N. Carvalho, and S. Pereira. 2023. "FluidHarmony: Defining an Equal-Tempered and Hierarchical Harmonic Lexicon in the Fourier Space." *Journal of New Music Research* 51 (2–3): 142–161. https://doi.org/10.1080/09298215.2023.2202641.

Bernardes, G., D. Cocharro, M. Caetano, C. Guedes, and M. E. Davies. 2016. "A Multi-Level Tonal Interval Space for Modelling Pitch Relatedness and Musical Consonance." *Journal of New Music Research* 45 (4): 281–294. https://doi.org/10.1080/09298215.2016.1182192.

Bharucha, J., and C. L. Krumhansl. 1983. "The Representation of Harmonic Structure in Music: Hierarchies of Stability as a Function of Context." *Cognition* 13 (1): 63–102.

Chiu, M. 2021. "Macroharmonic Progressions through the Discrete Fourier Transform: An Analysis from Maurice Duruflé's Requiem." *Music Theory Online* 27 (3). https://doi.org/10.30535/mto.27.3.1.

Domínguez, A. M. 2009. "Tonal Stability Modeling from Audio Chroma Features." PhD diss., Universitat Pompeu Fabra.

Dzemyda, G., O. Kurasova, and J. Žilinskas. 2013. *Multidimensional Data Visualization: Methods and Applications*. Springer Optimization and Its Applications 75. New York: Springer. https://doi.org/10.1007/978-1-4419-0236-8.

Granroth-Wilding, M. T. 2013. "Harmonic Analysis of Music Using Combinatory Categorial Grammar." PhD diss., University of Edinburgh.

Harasim, D., G. Affatato, and F. C. Moss. 2022. "midiVERTO: A Web Application to Visualize Tonality in Real Time." In *Mathematics and Computation in Music*, edited by M. Montiel, O. A. Agustín-Aquino, F. Gómez, J. Kastine, E. Lluis-Puebla, and B. Milam, 363–368. Lecture Notes in Computer Science. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-031-07015-0_31.

Harding, J. D. 2020. "Computer-Aided Analysis across the Tonal Divide: Cross-Stylistic Applications of the Discrete Fourier Transform." In *Music Encoding Conference*.

Krumhansl, C. L., and E. J. Kessler. 1982. "Tracing the Dynamic Changes in Perceived Tonal Organization in a Spatial Representation of Musical Keys." *Psychological Review* 89 (4): 334–368.

Laneve, S., L. Schaerf, G. Cecchetti, J. Hentschel, and M. Rohrmeier. 2023. "The Diachronic Development of Debussy's Musical Style: A Corpus Study with Discrete Fourier Transform." *Humanities and Social Sciences Communications* 10 (1): 1–13. https://doi.org/10.1057/s41599-023-01796-7.

Lewin, D. 1959. "Re: Intervallic Relations between Two Collections of Notes." *Journal of Music Theory* 3 (2): 298–301. https://doi.org/10.2307/842856.

Lieck, R., and M. Rohrmeier. 2020. "Modelling Hierarchical Key Structure with Pitch Scapes." In *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR 2020)*, 811–818. Montreal, Canada. https://doi.org/10.5281/zenodo.4245558.

McCreless, P. 1991. "Syntagmatics and Paradigmatics: Some Implications for the Analysis of Chromaticism in Tonal Music." *Music Theory Spectrum* 13 (2): 147–178. https://doi.org/10.2307/745896.

Moss, F. C., G. Affatato, and D. Harasim. 2022. "Phantom Curves: Scientific Discovery through Interactive Music Visualization." In *9th International Conference on Digital Libraries for Musicology*, edited by L. Pugin, 60–64. New York: Association for Computing Machinery. https://doi.org/10.1145/3543882.3543886.

Pereira, S., G. Bernardes, and J. O. Martins. 2025. "Qualia Motion in Fourier Space: Formalizing Linear, Nondirected, and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1." *Music Theory Spectrum*, advance article, mtaf016. https://doi.org/10.1093/mts/mtaf016.

Quinn, I. 2006. "General Equal-Tempered Harmony (Introduction and Part I)." *Perspectives of New Music* 44 (2): 114–158.

Quinn, I. 2007. "General Equal-Tempered Harmony: Parts 2 and 3." *Perspectives of New Music* 45 (1): 4–63.

Rohrmeier, M. 2011. "Towards a Generative Syntax of Tonal Harmony." *Journal of Mathematics and Music* 5 (1): 35–53. https://doi.org/10.1080/17459737.2011.573676.

Sapp, C. S. 2001. "Harmonic Visualizations of Tonal Music." In *Proceedings of the International Computer Music Conference*, 1:419–422. Havana, Cuba.

Sapp, C. S. 2005. "Visual Hierarchical Key Analysis." *Computers in Entertainment* 3 (4): 1–19. https://doi.org/10.1145/1095534.1095544.

Saussure, F. de, W. Baskin, P. Meisel, and H. Saussy. 2011. *Course in General Linguistics*. New York: Columbia University Press.

Temperley, D. 2007. "The Tonal Properties of Pitch-Class Sets: Tonal Implication, Tonal Ambiguity, and Tonalness." *Computing in Musicology* 15.

Viaccoz, C., D. Harasim, F. C. Moss, and M. Rohrmeier. 2023. "Wavescapes: A Visual Hierarchical Analysis of Tonality Using the Discrete Fourier Transform." *Musicae Scientiae* 27 (2): 390–427. https://doi.org/10.1177/10298649211034906.

Yust, J. 2016. "Special Collections." *Journal of Music Theory* 60 (2): 213–262. https://doi.org/10.1215/00222909-3651886.

Yust, J. 2019. "Stylistic Information in Pitch-Class Distributions." *Journal of New Music Research* 48 (3): 217–231. https://doi.org/10.1080/09298215.2019.1606833.